

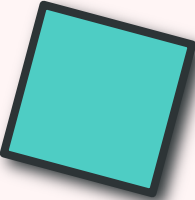
TECHNICAL REPORT

Nanbeige4-3B

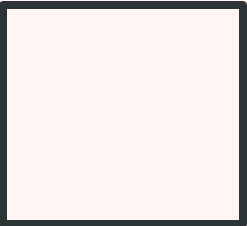
Exploring the Frontier of Small Language Models



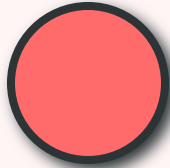
Nanbeige LLM Lab, Boss Zhipin



Pretrained on 23T Tokens • 30M+ Instructions • State-of-the-Art Performance



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The small language model revolution and key innovations

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Pre-Training: Data & Recipe

Data construction, quality assessment, and FG-WSD scheduler

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Evaluation & Results

Benchmark performance and WritingBench leaderboard



16 Slides • Comprehensive Technical Deep Dive

The Small Language Model Revolution

⚠ The Challenge

Large-scale language models (hundreds of billions to trillions of parameters) have demonstrated remarkable reasoning abilities but come with **substantial inference costs** and **high training expenses**, making deployment and research prohibitively expensive.

💡 The Solution

Nanbeige4-3B presents a **compact yet powerful alternative** that rivals much larger models across mathematics, scientific reasoning, human preference alignment, creative writing, and tool use—substantially outperforming many larger models.

🏆 **Key Achievement:** Outperforms Qwen3-8B and rivals Qwen3-14B despite being only 3B parameters

Performance vs Qwen3 Series

Base Models (Same SFT)

Nanbeige4-3B-Base >> Qwen3-8B-Base

Reasoning Models

Nanbeige4-3B-Thinking >> Qwen3-14B

Even outperforms Qwen3-32B on math & science!

Core Capabilities

- Mathematics
- Code
- Writing
- Science
- Tool Use
- Alignment

Model Overview & Key Innovations

Training at Scale

23T

Pre-training Tokens
High-quality curated data

30M+

SFT Instructions
Diverse fine-tuning data

Model Variants



Nanbeige4-3B-Base

Foundation model with strong initialization for downstream tasks



Nanbeige4-3B-Thinking

Reasoning-enhanced model with advanced post-training

Five Key Technical Innovations

1

Hybrid Data Filtering

Tagging + retrieval-based scoring

2

Fine-Grained WSD

Progressive data curriculum

3

Deliberative Generation

CoT reconstruction mechanism

4

Dual Preference Distillation

Token + sequence level optimization

5

Multi-Stage RL

Domain-specific reinforcement learning

Pre-Training: Data Construction Strategy

Comprehensive Data Collection

Web Pages

Clean HTML text extraction

Scholarly Articles

Academic papers & research

Books

Long-form literature

Source Code

Programming repositories

PDF Text Extraction Pipeline

Developed a **highly efficient PDF text-extraction pipeline** to strengthen the model's grasp of the human world, extracting clean text from complex document formats beyond simple HTML sources.

✓ Goal: High-quality, diverse data spanning multiple domains and formats

Synthetic Data Augmentation

15%

of Total Tokens

- Question-Answer Pairs
- Textbooks
- Lecture Notes
- Long CoT Samples

Final Corpus

23T

Total Tokens

High-quality, diverse training corpus

Data Quality Assessment: Two-Pronged Approach



Multi-Dimensional Tagging

Labeling Framework

Started with **60+ dimensions**, refined to **20 key dimensions** based on inter-dimension similarity and effectiveness

Key Dimensions Include

Knowledge Density

Reasoning Density

Text Fluency

💡 Key Finding: 0-9 scoring > binary 0-1 labeling

★ Content labels more predictive than format labels



Similarity-Based Scoring

Retrieval Infrastructure

Built a **retrieval database** with hundreds of billions of entries, supporting efficient hybrid text-based and vector-based retrieval

Seed Data Curation

Prioritize samples ranking high in quality tagging + ensure data provenance from reliable, authoritative sources

✓ Discovery: Retrieval on top of quality labeling enables more precise identification

Data Filtering Pipeline

Initial Corpus

Tens of T



Quality Filtered

12.5T



Up-sampled HQ

6.5T



Final Corpus

23T

Fine-Grained WSD: Revolutionizing Training Schedulers

FG-WSD Architecture

1 Warm-up Phase

Learning rate increases gradually

2 Multi-Stage Stable

Progressively refined data mixtures

3 Smooth Decay

Learning rate smoothly decreases

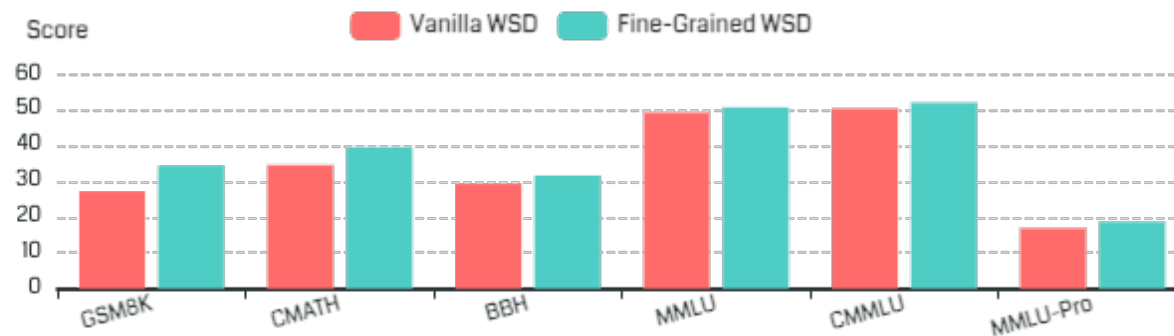
📈 Key Innovation: Prevents coupling between data ordering and LR changes

Preliminary Experiment (1B Model, 1T Tokens)

Vanilla WSD: Static 1:1 mixture of HQ and MQ data throughout

FG-WSD: Stage 1: 750B (1×HQ + 1×MQ) → Stage 2: 250B (HQ only)

Performance Gains: Vanilla WSD vs FG-WSD



Nanbeige4-3B-Base Training Stages

Warmup Stage

0.1T tokens | LR: 0 → 4.5e-4

Diversity-Enriched Stable

12.4T tokens | LR: 4.5e-4

High-Quality Stable

6.5T tokens | LR: 4.5e-4

Decay Stage (ABF)

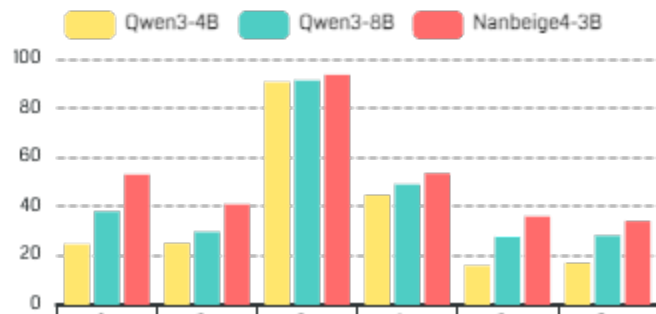
4T tokens | LR: 4.5e-4 → 1.5e-6

⚡ ABF method extends context length to 64K in decay stage

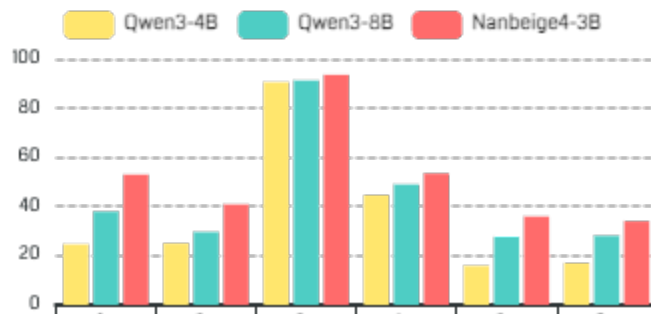
Base Model Evaluation: Post-SFT Performance

Evaluation Methodology: Fine-tuned Nanbeige4-3B-Base and Qwen3-Base series with **identical SFT pipelines** across three datasets (Nemotron, Ring-Lite, OpenThoughts). Each dataset: 500K instances, 2 epochs, identical hyperparameters.

Nemotron Dataset



Ring-Lite Dataset



OpenThoughts Dataset



🏆 **Key Finding:** Nanbeige4-3B-Base consistently outperforms Qwen3-8B-Base despite being half the size!

Post-Training Pipeline Overview

Complete Four-Stage Pipeline: From foundational reasoning skills through general capability expansion, knowledge distillation from flagship teacher model, to final reinforcement learning optimization across multiple domains.



Stage 1 Cold-Start SFT

Foundational Reasoning Skills



Stage 2 Overall SFT

General/Curriculum Learning



Stage 3 Distillation

Knowledge Transfer



Stage 4 Multi-Stage RL

Robust Reward System

→ Output: Nanbeige4-3B-Thinking with State-of-the-Art Performance

Cold Start SFT: Scaling Instructions to 30M+

Dataset Composition

Total Samples

30M+

Cleaned QA samples after filtering

Corpus Distribution

Mathematical Reasoning 50%

Scientific Reasoning 30%

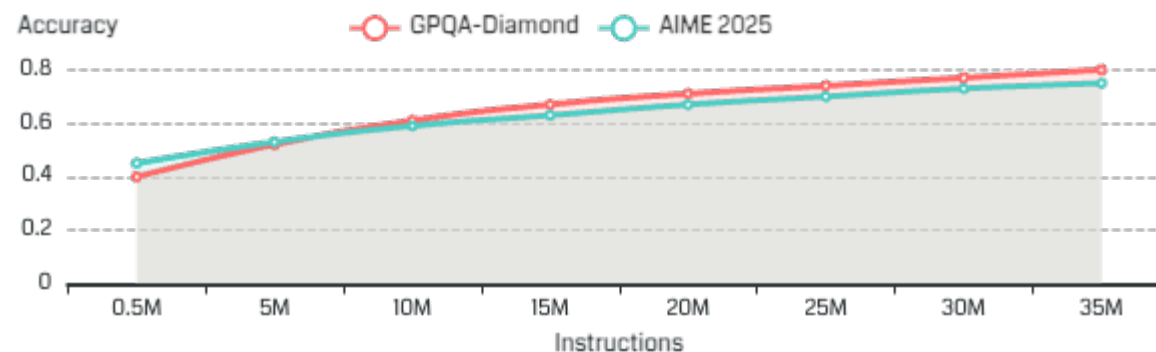
Code-Related Tasks 20%

📖 Context Length: 32K tokens

🎯 Objective: Strengthen CoT reasoning & structured responses

Key Finding: Scaling SFT Instructions

Contrary to recent reports suggesting hundreds of thousands of instructions suffice, our experiments reveal that **scaling to tens of millions** continues producing substantial improvements on challenging benchmarks without clear saturation.



AIME 2025 Performance

0.5M → 35M

Continuous improvement

GPQA-Diamond

0.40 → 0.80

2x accuracy gain

Overall SFT: Deliberative Generation & CoT Reconstruction

Overall SFT Data Mix (64K Context)



Two-Step Innovation Mechanism

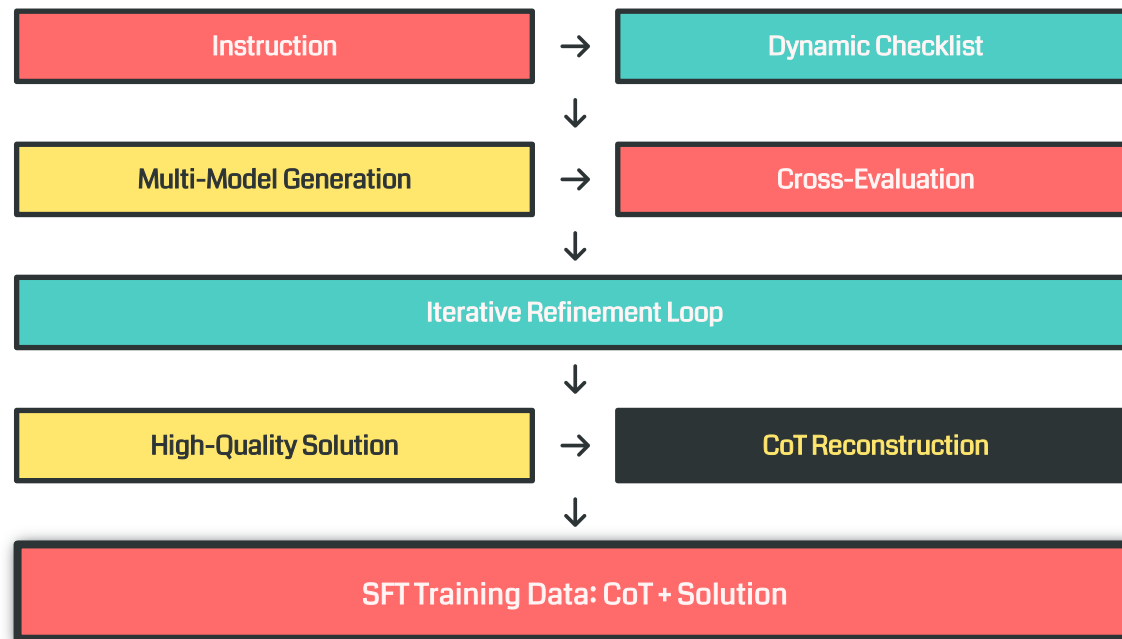
1 Solution Refinement

- Multi-dimensional evaluation checklist tailored to instruction type
- Criteria: correctness, completeness, consistency, executability, safety
- Iterative generate-review-revise cycles with teacher models

2 CoT Reconstruction

- Train a chain-completion model to restore reasoning
- Input: instruction + refined final solution
- Output: structured CoT + high-quality answer

Process Flow



📈 Arena-Hard v2: +16% improvement

Dual-Level Preference Distillation (DPD)

DPD Framework

Joint Training Objective

Token-Level Distillation

Match teacher probability distribution

Sequence-Level DPO

Maximize margin between pos/neg

Dual-Granularity = Better Alignment + Reasoning

Data Construction

+ Positive Samples

- Sample multiple responses from teacher
- Model-based scoring + rule filtering
- Select highest-scoring answers

- Negative Samples

- Sample from student model
- Auto evaluation + rule checks
- Retain significantly worse than positive

Optimization Objective

Joint Loss: Token-level distillation on **positive** samples + token-level on **negative** samples + sequence-level DPO margin constraint

Positive Token Distillation

Student matches teacher distribution

Negative Token Distillation

Reduce confident errors, boost alternatives

Performance Gains

~8%

AIME 2024/2025
Math reasoning

~10%

GPQA
Science reasoning

~30%

BFCL V4
Tool use

~8%

Arena-Style
Human preference

🔑 RL on top of DPD yields substantially larger gains than from SFT baseline

Multi-Stage Reinforcement Learning

Multi-Stage vs Single-Stage: Instead of mixed-corpus training, we organize RL into **multiple stages**, each focused on a single domain. This allows concentrated capacity on domain-specific skills while benefiting from cross-domain transfer.



Phase 1
STEM RL

Agentic Verifier



Phase 2
Coding RL

Synthetic Test Functions



Phase 3
Human Pref RL

Pairwise Reward Model

🔴 On-Policy Data Filtering

Before each RL stage, use previous stage model to compute **avg@16 accuracy**. Retain samples with **10-90% pass rate** — neither trivial nor unsolvable.

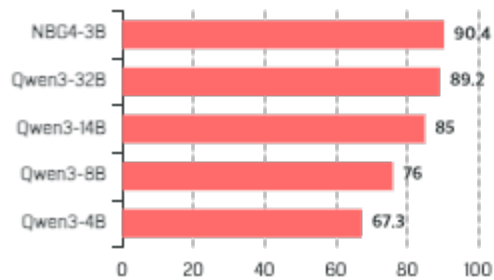
⚙️ Training Enhancements

On-policy GRPO with KL penalty removal and truncated sequence loss masking (following DAPO insights) for stabilized training.

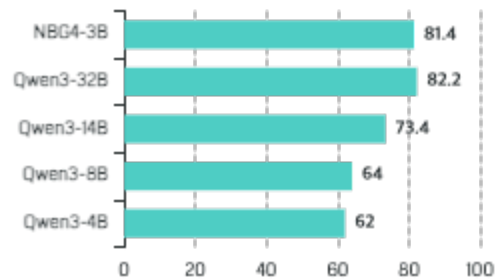
Final Evaluation: Benchmark Results

Evaluation Setup: Sampling temperature 0.6, top-p 0.95, max generation 64K tokens. Multiple evaluation runs with averaged scores (avg@8 for AIME, avg@3 for others).

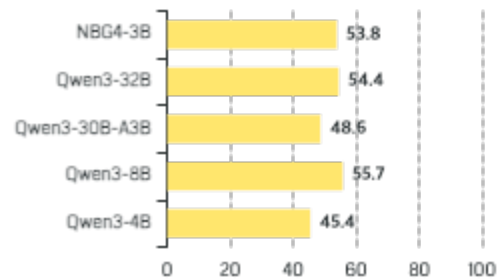
Mathematical Reasoning



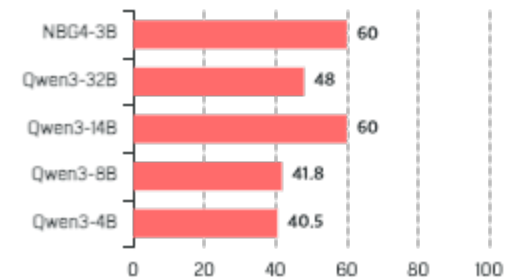
Scientific Reasoning



Tool Use & Coding



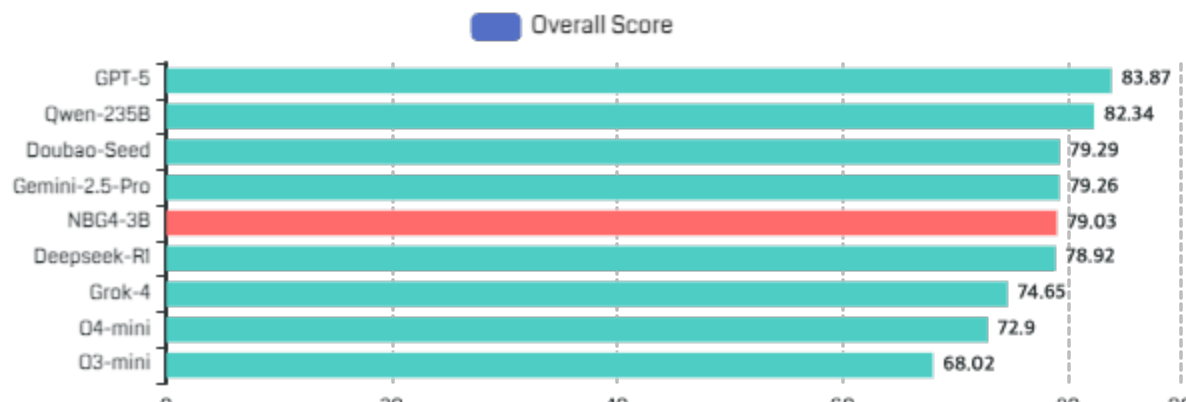
Human Preference



🏆 Nanbeige4-3B sets new SOTA on AIME & GPQA, outperforming models with 10x more parameters!

WritingBench: Creative Writing Excellence

WritingBench Leaderboard (Nov 2025)



★ Nanbeige4-3B-Thinking ranks among top models including GPT-5 and Deepseek-R1!

Seven Writing Categories

Academic & Engineering	80.77
Finance & Business	82.37
Politics & Law	77.41
Literature & Arts	76.05
Education	80.81
Advertising & Marketing	79.03

Overall Score

79.03

5th place overall

CONCLUSION

Nanbeige4-3B: Small Model, **Massive** Impact

A 3-billion-parameter model that redefines small-scale potential through innovation in data curation and training paradigm. Despite its modest size, Nanbeige4-3B **outperforms Qwen3-8B and Qwen3-14B** on challenging benchmarks, rivaling Deepseek-R1 on WritingBench.



New SOTA

AIME & GPQA benchmarks



10x Efficiency

vs larger parameter models



Open Source

huggingface.co/Nanbeige

Future Directions

Autonomous Software Engineering • Deep-Research Agents • Cross-Scenario Tool Use

Thank You | Questions Welcome